A one-sided endpoint complex interpolation result for conditioned noncommutative martingale Hardy and BMO spaces

Automatic Math Discovery Run `fa_banach_001`

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Status

This is a candidate partial result for the open problem in Randrianantoanina's paper [1]. It proves one inclusion in the range where the target exponent is > 1. It does not prove the reverse inclusion, and it does not settle the range where the target exponent is ≤ 1.

Source problem

The source paper proves that, for finite indices $0 < p, q < \infty$,

$$[\mathsf{h}_p^c(\mathcal{M}), \mathsf{h}_q^c(\mathcal{M})]_\theta = \mathsf{h}_r^c(\mathcal{M}), \qquad \frac{1}{r} = \frac{1-\theta}{p} + \frac{\theta}{q}.$$

Immediately after that theorem, the paper asks whether the corresponding statement remains valid when $0 < p < 1$ and the endpoint couple $(\mathsf{h}_p^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M}))$ is used.

> From combining (3.12) and (3.13), we obtain the desired equivalence. $\square$
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> When $0 < p < 1$, we do not know if the corresponding statement to Theorem 3.16 remains valid if the interpolation couple $(\mathsf{h}_p^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M}))$ is used. Since reiteration theorem is not available for complex interpolations of quasi-Banach spaces, in general, this consideration is independent of Theorem 3.16. We leave this as an open problem.
>
> The same method of proofs can be applied to conditioned spaces and spaces of adapted se-

Thus the desired full endpoint identity would be

$$[\mathsf{h}_p^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M})]_\theta = \mathsf{h}_r^c(\mathcal{M}), \qquad \frac{1}{r} = \frac{1-\theta}{p}.$$

Proof intuition

The source already proves the analogous one-sided inclusion for the larger capital Hardy/BMO couple by using Wolff's interpolation lemma at the level of inclusions. The same interpolation geometry also applies to the conditioned column spaces. The finite-index complex theorem gives $[\mathsf{h}_p^c, \mathsf{h}_r^c]_\eta = \mathsf{h}_1^c$, while the known Banach-range endpoint theorem of Bekjan–Chen–Perrin–Yin gives $[\mathsf{h}_1^c, \mathsf{bmo}^c]_\phi = \mathsf{h}_r^c$ when $r > 1$. Wolff's inclusion lemma then splices these two identities and yields the inclusion from $[\mathsf{h}_p^c, \mathsf{bmo}^c]_\theta$ into h_r^c. The argument deliberately uses only the inclusion half of Wolff's mechanism, because full complex reiteration is not available in this quasi-Banach endpoint setting.

Theorem 1. *Let $\mathcal{M}$ be as in [1]: a semifinite von Neumann algebra with an increasing filtration admitting trace-preserving conditional expectations. Let $0 < p < 1$ and $0 < \theta < 1$, and put*

$$\frac{1}{r} = \frac{1-\theta}{p}.$$

If $r > 1$, then

$$[\mathsf{h}_p^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M})]_\theta \hookrightarrow \mathsf{h}_r^c(\mathcal{M})$$

continuously.

Proof. Choose $\eta \in (0,1)$ so that

$$1 = \frac{1-\eta}{p} + \frac{\eta}{r}.$$

This is possible because $p < 1 < r$. By the finite-index complex interpolation theorem of [1, Theorem 3.16], applied to the two finite exponents p and r, we have

$$[\mathsf{h}_p^c(\mathcal{M}), \mathsf{h}_r^c(\mathcal{M})]_\eta = \mathsf{h}_1^c(\mathcal{M})$$

with equivalent quasi-norms.

Set

$$\phi = 1 - \frac{1}{r}.$$

Since $r > 1$, the Banach-range endpoint theorem of Bekjan–Chen–Perrin–Yin [2, Theorem 4.4], with the order of the couple reversed, gives

$$[\mathsf{h}_1^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M})]_\phi = \mathsf{h}_r^c(\mathcal{M}).$$

We also need the intersection condition in Wolff's inclusion lemma. If $x \in \mathsf{h}_p^c(\mathcal{M}) \cap \mathsf{bmo}^c(\mathcal{M})$, then the real endpoint identity

$$(\mathsf{h}_p^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M}))_{\alpha,s} = \mathsf{h}_s^c(\mathcal{M}), \qquad \frac{1}{s} = \frac{1-\alpha}{p},$$

from [1, Theorem 3.5] implies, by the elementary K-functional estimate for elements of the intersection, that

$$x \in \mathsf{h}_1^c(\mathcal{M}) \cap \mathsf{h}_r^c(\mathcal{M}).$$

Hence

$$\mathsf{h}_p^c(\mathcal{M}) \cap \mathsf{bmo}^c(\mathcal{M}) \subset \mathsf{h}_1^c(\mathcal{M}) \cap \mathsf{h}_r^c(\mathcal{M}).$$

Apply Wolff's complex interpolation lemma in the inclusion form used in [1, Corollary 3.19] to

$$B_1 = \mathsf{h}_p^c(\mathcal{M}), \qquad B_2 = \mathsf{h}_1^c(\mathcal{M}), \qquad B_3 = \mathsf{h}_r^c(\mathcal{M}), \qquad B_4 = \mathsf{bmo}^c(\mathcal{M}).$$

The two interpolation identities above give the hypotheses $[B_1, B_3]_\eta \subset B_2$ and $[B_2, B_4]_\phi \subset B_3$. Therefore

$$[B_1, B_4]_\Theta \subset B_3, \qquad \Theta = \frac{\phi}{1-\eta+\eta\phi}.$$

It remains only to check the parameter. Write $a = 1/p$ and $b = 1/r$. Then

$$\eta = \frac{a-1}{a-b}, \qquad \phi = 1 - b,$$

and hence

$$\Theta = \frac{1-b}{1-\eta b} = \frac{1-b}{1-\frac{b(a-1)}{a-b}} = \frac{a-b}{a} = 1 - \frac{p}{r} = \theta,$$

because $1/r = (1-\theta)/p$. Thus

$$[\mathsf{h}_p^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M})]_\theta \subset \mathsf{h}_r^c(\mathcal{M}),$$

as claimed. □

Limitations

The proof gives only the inclusion

$$[\mathsf{h}_p^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M})]_\theta \subset \mathsf{h}_r^c(\mathcal{M})$$

when $r > 1$. It does not give

$$\mathsf{h}_r^c(\mathcal{M}) \subset [\mathsf{h}_p^c(\mathcal{M}), \mathsf{bmo}^c(\mathcal{M})]_\theta.$$

Obtaining that reverse inclusion would require a new endpoint construction or a full quasi-Banach complex reiteration principle. This is precisely the obstruction identified in the source paper. The argument also does not cover the case $r \le 1$, since the required intermediate Banach-range theorem $[\mathsf{h}_1^c, \mathsf{bmo}^c]_\phi = \mathsf{h}_r^c$ is only available for $r > 1$.

Verification notes

The proof is symbolic and uses no numerical computation. The parameter calculation was checked algebraically in the proof. The main review point is the use of Wolff's interpolation lemma in the same quasi-Banach inclusion form invoked by [1, Corollary 3.19].

Novelty check

The run's lightweight indexes were searched for arXiv:2108.06341, title terms, and core phrases such as h_p^c, bmo^c, and complex interpolation; no prior packet or attempt was found. Web searches for exact and close variants of the endpoint question found the source paper, the Bekjan–Chen–Perrin–Yin Banach-range theorem [2], the 2022 real-method h_∞^c sequel, and the 2024 capital-H real-method sequel, but no later full solution of this conditioned BMO complex endpoint.

References

[1] N. Randrianantoanina, *Interpolation between noncommutative martingale Hardy and BMO spaces: the case* $0 < p < 1$, arXiv:2108.06341.

[2] T. N. Bekjan, Z. Chen, M. Perrin, and Z. Yin, *Atomic decomposition and interpolation for Hardy spaces of noncommutative martingales*, arXiv:0906.4437.

[3] T. Wolff, *A note on interpolation spaces*, Harmonic Analysis (Minneapolis, Minn., 1981), Lecture Notes in Math. 908, Springer, 1982, 199–204.